

1 Simulation study on $\mathbb{S}^1$

We evaluate frequentist risk performance of the proposed denoising estimators on the unit circle ($\mathbb{S}^1$). Let $\Theta \in \mathbb{S}^1$ denote a latent signal and $X \in \mathbb{S}^1$ its noisy observation generated from the isotropic Riemannian normal model

$$X \mid \Theta \sim \mathcal{N}_{\mathbb{S}^1}(\Theta, \sigma^2),$$

implemented via `random_riemannian_normal` available in `Geomstats`. Performance of a denoiser δ is measured using squared geodesic loss,

$$\mathcal{R}(\delta) := \mathbb{E}[d^2(\delta(X), \Theta)],$$

where $d(\cdot, \cdot)$ is the geodesic distance on $\mathbb{S}^1$.

To probe robustness across increasing multimodality, we consider four sampling schemes G for Θ , namely: (i) one-modal, a single Riemannian Gaussian mode with variance $\tau^2 = 0.05$; (ii) two-modal, a two-component Riemannian Gaussian mixture with separated modes and component variance $\tau^2 = 0.04$; (iii) three-modal, a three-component separated mixture with component variance $\tau^2 = 0.03$; and (iv) four-modal, a four-component separated mixture with component variance $\tau^2 = 0.02$. For visualization we display a circular kernel density estimate of each G on a polar plot.

We compare the Naïve baseline $\delta_N(X) = X$, corresponding to no denoising, with the oracle Tweedie denoiser δ_T and the oracle Bayes estimator δ_B , which in the (rather exceptional) case of the circle can be computed in closed form. These oracle procedures are computed by Monte Carlo integration using independent sampling from the true G and therefore serve as benchmarks for the best achievable behavior under the model (bypassing density/score estimation). Concretely, δ_T is based on approximating the conditional expectation $\mathbb{E}[\log_X \Theta \mid X]$, while δ_B approximates the Bayes rule by rejection sampling.

We first examine how risk varies with the noise variance σ^2 . For each target G , we evaluate the Monte Carlo risk

$$\widehat{\mathcal{R}}(\delta) := \frac{1}{n} \sum_{i=1}^n d^2(\delta(X_i), \Theta_i)$$

with $n = 500$. We consider increasing values of σ^2 on a grid in $[0.005, 0.25]$ and repeat the experiment across $N_{\text{MC}} = 3$ Monte Carlo replications. The output is plotted with mean risk and 68% confidence intervals across replications.

This experiment highlights the dependence of denoising gains on the underlying signal distribution G . As expected, for unimodal targets the oracle procedures substantially improve over δ_N as σ^2 increases. As multimodality increases, the gap between δ_T and δ_B becomes more visible at moderate-to-large noise levels, reflecting the increasing difficulty of denoising under highly non-uniform, multi-peak structure.

The second experiment studies the finite-sample excess risk induced by estimating the denoiser from data relative to its oracle counterpart. Concretely, for

each G , each σ^2 , and each training sample size $N \in \{50, 100, 250, 500\}$, we draw an independent test set, compute the denoiser $\hat{\delta}_T(\cdot; \rho)$ using fixed complexity ($M = 7$) and a grid of regularization parameters ρ , and select ρ^* by minimizing the empirical test risk. We then compute the oracle Tweedie benchmark on the same test set and display the difference

$$\Delta(N; \sigma^2, G) := \hat{\mathcal{R}}_{\text{test}}\left(\hat{\delta}_T(\cdot; \rho^*)\right) - \hat{\mathcal{R}}_{\text{test}}(\delta_T)$$

across $N_{\text{MC}} = 10$ replications. We plot Δ versus N on a log (x)-scale, with separate curves for each σ^2 .

The excess risk decays with N , consistent with improved stability of the estimated score/denoiser and more reliable tuning as the training sample size increases. The decay is slower and the gap larger for higher σ^2 and for more multimodal targets, where accurate estimation is more challenging.

In summary, the $\mathbb{S}^1$ simulations demonstrate: (i) substantial oracle gains over the Naïve estimator when G is concentrated; (ii) increasing multimodality makes denoising harder, amplifying the gap between δ_T and δ_B at moderate-to-large noise; and (iii) the excess risk of empirically tuned procedures over the oracle benchmark decreases with training sample size but is most pronounced for high noise and highly multimodal regimes.

Simulation study on $\mathbb{S}^2$

We evaluate frequentist risk performance of the proposed denoising estimators on the two-dimensional sphere ($\mathbb{S}^2$). Let $\Theta \in \mathbb{S}^2$ denote a latent signal and $X \in \mathbb{S}^2$ its noisy observation generated from the isotropic Riemannian normal model

$$X \mid \Theta \sim \mathcal{N}_{\mathbb{S}^2}(\Theta, \sigma^2),$$

implemented via `random_riemannian_normal` available on `Geomstats` python module. Performance of a denoiser δ is measured using squared geodesic loss, i.e.

$$\mathcal{R}(\delta) := \mathbb{E}[d^2(\delta(X), \Theta)],$$

where $d(\cdot, \cdot)$ is the geodesic distance on $\mathbb{S}^2$.

To probe robustness across qualitatively different signal structures, we consider four sampling schemes G for Θ , each implemented through `get_G_class('S2', sampler, name, params)`:

- Uniform: $\Theta \sim \text{Unif}(\mathbb{S}^2)$.
- One-modal: a single concentrated mode, i.e. a Gaussian Riemannian sampler with variance $\tau^2 = 0.05$.
- Four-modal: a Gaussian Riemannian mixture sampler with four separated modes ($\tau^2 = 0.01$ for each).
- Equatorial: mass concentrated around the equator geodesic.

These four targets span regimes from fully diffuse (uniform) to strongly non-uniform and highly structured (multimodal/equatorial). For visualization we display a kernel density estimate of each G on a Mollweide projection.

We a Naïve baseline $\delta_N(X) = X$, corresponding to no denoising, with the oracle denoiser δ_T .

The latter is “oracle” in the sense that it have access to independent sampling from the true G and therefore serve as benchmarks for the best achievable behavior under the model, and does not go through estimation of the density/score, but rather by the (approximat) conditional expectation $\mathbb{E}[\log_X \Theta \mid X]$ computed by Monte Carlo integration.

We first examine how risk varies with the noise variance σ^2 . For each target G , we evaluate the monte carlo risk

$$\widehat{\mathcal{R}}(\delta) := \frac{1}{n} \sum_{i=1}^n d^2(\delta(X_i), \Theta_i)$$

for some large n , say $n = 500$. We consider increasing values of $\sigma^2 \in \{0.005, 0.01, 0.05, 0.10, 0.15, 0.20, 0.25\}$ and repeat the experiment across $N_{\text{MC}} = 20$ Monte Carlo replications. The output is plotted with mean risk and 68% confidence intervals across replications.

This experiment quantifies the expected ordering between the no-denoising baseline δ_N and the oracle procedure as noise increases, and highlights the dependence of the potential denoising gains on the underlying signal distribution G . As expected, under the uniform target (no structure to exploit) the naïve and oracle denoisers are nearly indistinguishable, whereas for structured targets (modal or equatorial) the oracle denoisers leverage concentration and exhibit substantially smaller risk, especially at moderate-to-large σ^2 .

The second experiment studies the finite-sample excess risk induced by estimating the denoiser from data and its oracle version. Concretely, for each G , each σ^2 , and each training sample size $N \in \{50, 100, 250, 500, 1000\}$, we draw an independent sample $\{\Theta_i\}_{i=1}^N$ and noisy X_i ; we compute the denoiser $\hat{\delta}_T(\cdot; \rho)$ using with fixed complexity ($M = 7$) and cross-validated regularisation ρ ; we compute the oracle and empirical Tweedie risk on the sets, and display the Monte Carlo distribution of the difference

$$\Delta(N; \sigma^2, G) := \widehat{\mathcal{R}}_{\text{test}}(\hat{\delta}_T(\cdot; \rho^*)) - \widehat{\mathcal{R}}_{\text{test}}(\delta_T),$$

between the two risks across 20 replications. We plot Δ versus N on a log (x)-scale, with separate curves for each σ^2 .

The curves decay with N , consistently with the theory, since larger sample size improves density/score estimation and stabilizes tuning

In summary, the $\mathbb{S}^2$ simulations demonstrate: (i) oracle denoisers can yield substantial risk reductions relative to δ_N when G is structured; (ii) the excess risk of empirically tuned procedures over their oracle counterpart decreases with training sample size and is most visible in high-noise, structured regimes.